



ExpressOrgChart Suite v1.2

Manual

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Getting started with ExpressOrgCharts

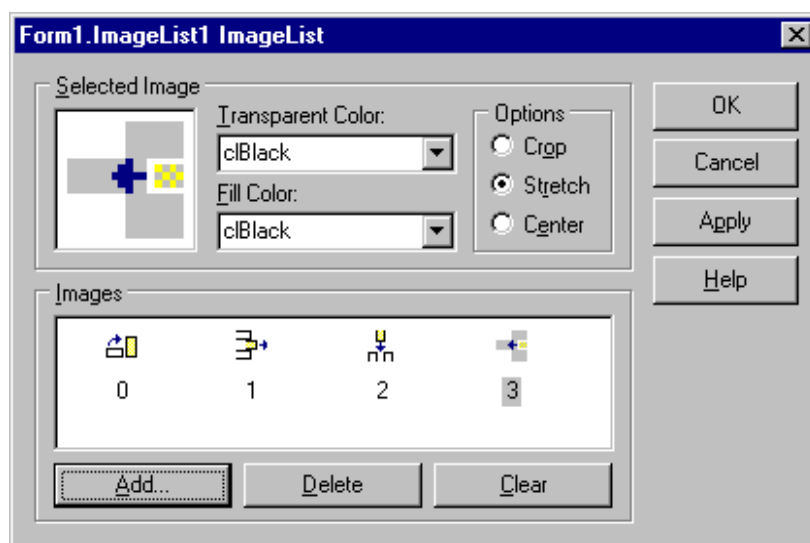
Upon the installation of the ExpressOrgChart package, your component palette will include an ExpressOrgCharts page with two components:

dxOrgChart: Non-Data Aware version of the ExpressOrgChart.

dxDbOrgChart: Data Aware version of the ExpressOrgChart.

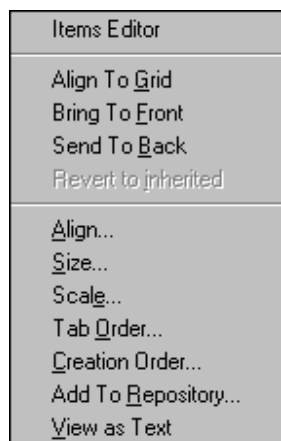
Using the non-data aware version of ExpressOrgChart

1. Drop a dxOrgChart component onto a form.
2. If you wish to associate images with individual chart nodes, drop a standard ImageList component onto the form and add all required images using the Image List Editor.

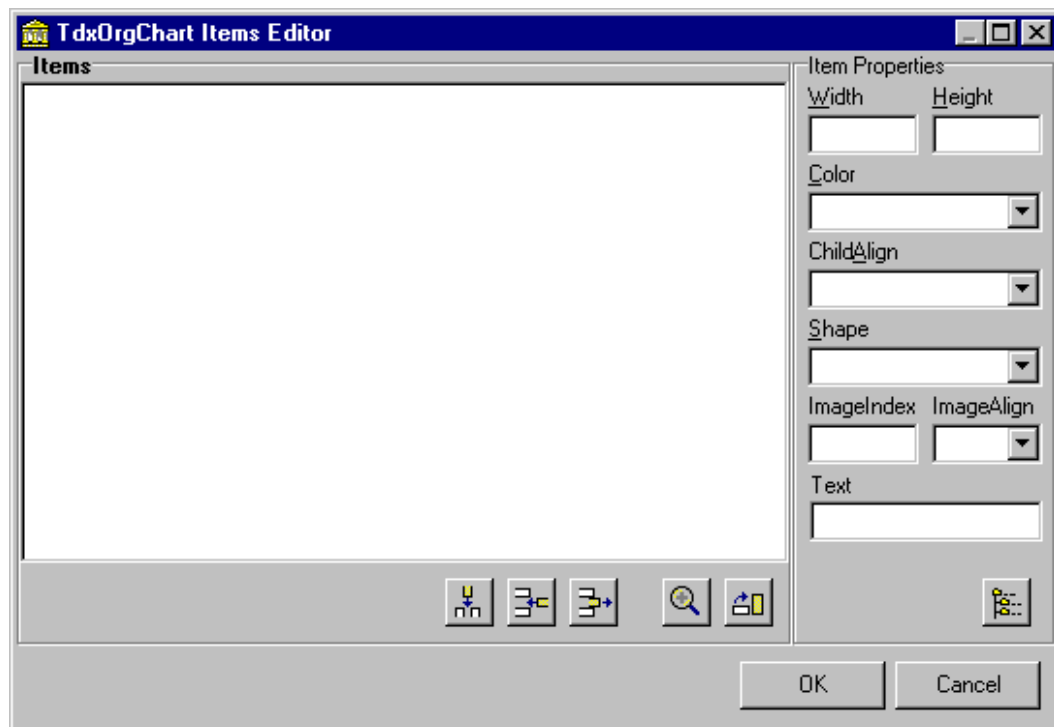


3. Using the Object Inspector, determine individual properties such as Alignment, Border Style, Color, etc. If tree nodes are dynamically created in the process of program work, then dxOrgChart is ready for use.

4. To create a data structure at design-time, call the component editor by double clicking the dxOrgChart component or right click within the component and select the Items Editor menu item to activate the Items Editor dialog window.



5. Using the Org Chart Item Editor window, you can begin to build the hierarchy and layout of the chart.



A. To insert a new node, press the Insert New Item command button.



B. To insert a sub node, first select the parent node to which the new sub node will belong then press the Insert New Sub Item command button.



C. To delete a node, first select it using your mouse then press the Delete Item command button.



D. If your node structure exceeds the boundaries of the chart client area, press the Zoom command button to resize it within the Org Chart Item Editor window.

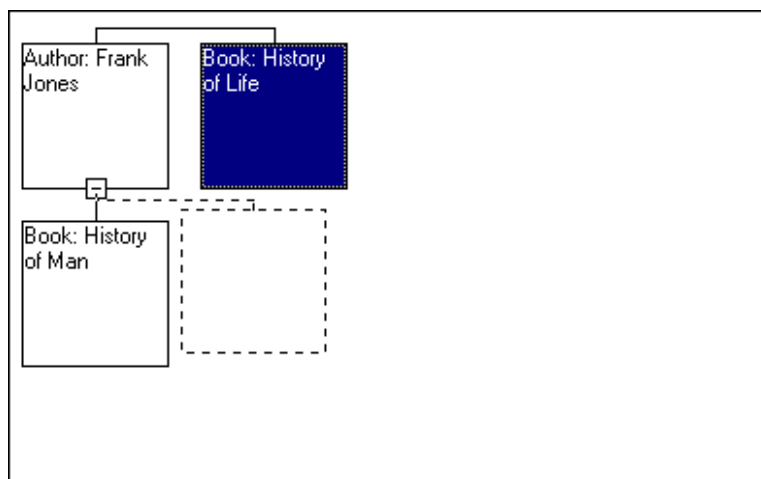


E. To change the rotation of the chart by a 90 degree angle, use the rotate on/off command button.



F. To set the properties of individual nodes within the chart, use the Item Properties pane of the Item Editor. Once you have determined the properties of an individual parent node, you can copy the "template" to all children by pressing the Set Properties for all children command button located in the lower right portion of the Item Editor window.

G. Using Drag and Drop, you can reorganize the chart nodes as your need require.



H. Set additional properties of the OrgChart using the Object Inspector.

I. Press the OK command button to commit your changes and return to the application form.

Using the data aware version of ExpressOrgChart

The data aware version of the ExpressOrgChart is suited for self-referenced tree structures. In general terms, in order to use the DBOrgChart, developers must assign a KeyField, ParentField, and a Text Field.

Note: Parent and KeyFields must be of the same type.

Additionally, developers may assign an OrderField to the OrgChart so as to manage the ordering of node objects within the tree hierarchy.

1. Drop a dxDbOrgChart component onto a form.
2. If you wish to associate images with individual chart nodes, drop a standard ImageList component onto the form and add all required images using the Image List Editor.
3. In order to activate the dxDbOrgChart, it is necessary to connect it to your database. Using the appropriate access components (TTable, TQuery, TDataSource) create links to your data.
4. Set the DataSource property of the dxDbOrgChart to the DataSource component on your form.
5. Using the Object Inspector, set the KeyFieldName and ParentFieldName properties and optional field names – TextFieldName and ImageFieldName.
6. Run your application to see the data represented within the OrgChart.

Options

Both components provide a wide set of possibilities to easily add end user functionality without writing code:

- ?? Setting standard width and height of tree nodes (DefaultNodeWidth and DefaultNodeHeight properties)
- ?? Setting thickness of contour and connecting lines (LineWidth property)
- ?? Setting distance between the neighboring nodes horizontally and vertically (IndentX and IndentY properties)
- ?? Different modes of location and justification of the text inside the nodes and also text editing (EditMode property)
- ?? Autoscaling (Zoom property)
- ?? Rotated display (Rotated property)

- ?? Embedded Drag & Drop interface with dynamic display of the dragging node (ocCanDrag and ocShowDrag in Options modes)
- ?? Possibility to add and delete nodes by Ins, Ctrl-Ins and Del keys (ocInsDel in Options mode)
- ?? 3-Dimensional display of nodes (ocRect3D in Options mode)
- ?? Animated expansion and collapse of nodes (ocAnimate in Options mode)

TdxCustomOrgChart

Unit

DxOrgChr

Description

TdxCustomOrgChart is an abstract class for displaying and editing the hierarchical tree data structure. It contains the base protected properties and methods published in their direct descendants: TdxOrgChart ? TdxDbOrgChart.

TdxCustomOrgChart Properties

BorderStyle	protected/read/write
Count	public/read/write
DefaultImageAlign	protected/read/write
DefaultNodeHeight	protected/read/write
DefaultNodeWidth	protected/read/write
Editing	public/read/write
EditMode	protected/read/write
Images	protected/read/write
IndentX	protected/read/write
IndentY	protected/read/write
Items	public/read/write
LeftEdge	public/read/write
LineWidth	protected/read/write
Options	protected/read/write
Rotated	protected/read/write
Selected	public/read/write
TopEdge	public/read/write
Zoom	public/read/write

BorderStyle property

read/write

BorderStyle determines whether the component has a border.

```
TBorderStyle = (bsNone, bsSingle);  
property BorderStyle: TBorderStyle;
```

Description

Set BorderStyle to specify a border frame around the component. If BorderStyle=bsSingle, then the component has a border, if it is bsNone, then it has no border. If Ctl3D=True, then the border is drawn with a three-dimensional effect.

Count property

read/write

Count represents the number of nodes in a tree structure.

```
property Count: Integer;
```

Description

Use Count to determine the total amount of tree nodes owned by the component. Count alters by deleting or adding nodes.

See also:

TdxOcNode.Count

DefaultImageAlign property**read/write**

DefaultImageAlign is the default value for TdxOcNode.ImageAlign property.

```
TdxOcImageAlign = (iaNone, iaLT, iaLC, iaLB, iaRT, iaRC, iaRB,  
    iaTL, iaTC, iaTR, iaBL, iaBC, iaBR);  
property DefaultImageAlign: TdxOcImageAlign;
```

Description

Set DefaultImageAlign to specify the standard location of tree node's images. When a new node is added to the tree, its ImageAlign property is set to this value.

See Also:

TdxCustomOrgChart.Images, TdxOcNode.ImageAlign, TdxOcNode.ImageIndex

DefaultNodeHeight property**read/write**

DefaultNodeHeight is the default height for tree nodes.

```
property DefaultNodeHeight: Word;
```

Description

The DefaultNodeHeight property determines the standard height of nodes in pixels. This height is adopted by default for all nodes having the Height property equaling 0. If the component does not have an image list, the minimal value of DefaultNodeHeight is 16, otherwise it is equal to the doubled height of the image.

Setting a new value to the DefaultNodeHeight will change the height of all nodes, except those nodes, whose Height property was explicitly set to a nonzero value.

See Also:

TdxCustomOrgChart.DefaultNodeWidth, TdxOcNode.High

DefaultNodeWidth property**read/write**

DefaultNodeWidth is the default width for tree nodes.

```
property DefaultNodeWidth: Word;
```

Description

The DefaultNodeWidth property determines the standard width of nodes in pixels. This width is adopted by default for all nodes having the Width property equaling 0. If the component does not have an image list, the minimal value of DefaultNodeWidth is 16, otherwise it is equal to the doubled width of the image.

Setting a new value to the DefaultNodeWidth will change the width of all nodes, except those nodes, whose Width property was explicitly set to a nonzero value.

See Also:

TdxCustomOrgChart.DefaultNodeHigh, TdxOcNode.Width

Editing property**read/write**

EditMode determines the output and text editing modes.

Type:

```
TdxOcEditMode = set of (emLeft, emCenter, emRight, emVCenter,
```

```
emWrap, emUpper, emLower, emGrow);
property EditMode: TdxOcEditMode;
```

Description

Set the EditMode flags to determine how text will be displayed within tree nodes. Additionally, EditMode sets specific properties of the in-place text editor.

Value	Meaning
emLeft	Text is left-justified in the client area of a node.
emCenter	Text is centered within a node.
emRight	Text is right-justified in the client area of a node.
emVCenter	Text is vertically centered within a node. This mode works successfully with single lines only.
emWrap	Text automatically wraps. When necessary, if this property is set to False, the width of a node increases to the text size.
emUpper	During editing, letters are converted into upper case.
emLower	During editing, letters are converted into lower case.
emGrow	If necessary, the height of nodes grow to the size of appropriate text.

See Also:

*TdxCustomOrgChart.BorderStyle, TdxCustomOrgChart.CreateEditor,
TdxCustomOrgChart.HideEditor, TdxCustomOrgChart.ShowEditor*

EditMode property

[read/write](#)

EditMode determines the output and text editing modes.

Type:

```
TdxOcEditMode = set of (emLeft, emCenter, emRight, emVCenter,
emWrap, emUpper, emLower, emGrow);
```

```
property EditMode: TdxOcEditMode;
```

Description

Set the EditMode flags to determine how text will be displayed within tree nodes. Additionally, EditMode sets specific properties of the in-place text editor.

Value	Meaning
emLeft	Text is left-justified in the client area of a node.
emCenter	Text is centered within a node.
emRight	Text is right-justified in the client area of a node.
emVCenter	Text is vertically centered within a node. This mode works successfully with single lines only.
emWrap	Text automatically wraps. When necessary, if this property is set to False, the width of a node increases to the text size.
emUpper	During editing, letters are converted into upper case.
emLower	During editing, letters are converted into lower case.
emGrow	If necessary, the height of nodes grow to the size of appropriate text.
emGrow	If necessary the height of the node grows to the text size.

See Also:

*TdxCustomOrgChart.CreateEditor, TdxCustomOrgChart.HideEditor,
TdxCustomOrgChart.ShowEditor*

Images property**read/write**

Determines which image list is associated with the component.

property Images: TImageList;

Description

Use Images to specify the TImageList component containing an image list for all TdxOrgChart nodes. The TdxOcNode.ImageIndex property sets an image from this list.

Note: When Images is set to a non Null value, the DefaultImageAlign property, if set originally to iaNone, automatically converts into iaLT.

See Also:

TdxCustomOrgChart.DefaultImageAlign, TdxOcNode.ImageAlign, TdxOcNode.ImageIndex

IndentX property**read/write**

IndentX is the width of the blank space separating nodes horizontally.

property IndentX: Word;

Description

Use IndentX to determine by what amount adjacent nodes are separated.

See Also:

TdxCustomOrgChart.IndentY

IndentY property**read/write**

IndentY is the height of the blank space separating nodes vertically.

property IndentY: Word;

Description

Use IndentY to determine by what amount child nodes are separated from their parents.

See Also:

TdxCustomOrgChart.IndentX

Items property**read/write**

Items is an indexed array of the top-level tree nodes managed by the component.

property Items[Index:Integer]: TdxOcNode;

Description

Use Items to access a top-level node by its index. The first node has an index of 0, the second an index of 1, etc. Items is a default property for TdxCustomOrgChart, so Chart.Items[I] is equal to Chart[I].

Note: TdxCustomOrgChart.Items allows access to top-level nodes only, having no parents. In order to access their children you must use the TdxOcNode.Items property.

See Also:*TdxOcNode.Items***LeftEdge property****read/write**

LeftEdge is the position of a window's left edge relative to the component's left edge.

property Items[Index:Integer]: TdxOcNode;

Description

Read LeftEdge to determine the area of the component which is visible. Set LeftEdge to change this visible area. When LeftEdge is changed, the window scrolls horizontally.

See Also:*TdxCustomOrgChart.SetLeftTop, TdxCustomOrgChart.TopEdge***LineWidth property****read/write**

LineWidth determines the width of all drawn lines.

property LineWidth: Word;

Description

Use LineWidth to set the width of lines for drawn nodes and the connections between them.

Note: In Zoom mode, LineWidth always returns 1, regardless of values set.

Options property**read/write**

Options determines the different runtime properties of the component.

Type:

TdxOcOptions = set of (ocSelect, ocFocus, ocButtons, ocDbClick, ocEdit, ocCanDrag, ocShowDrag, ocInsDel, ocRect3D, ocAnimate);

property Options: TdxOcOptions;

Description

Set the flags in Options to determine display features, as well as keyboard and mouse support for your component.

Value	Meaning
ocSelect	Selected item is marked by a focus rectangle.
ocFocus	The focused item is marked by clHighlight color.
ocButtons	Nodes with hildren are displayed with [+] and [-] buttons. In Zoom mode, only [+] buttons are displayed.
ocDbClick	Double click opens and closes the selected node.
ocEdit	Allows text editing in nodes.
ocCanDrag	The component starts Drag & Drop on dragging a node . When ocCanDrag = False the application may still activate Drag & Drop by calling BeginDrag.
ocShowDrag	Displays the location of a node and its connection with the potential parent during node drag operations.
oclnsDel	Inserts and deletes nodes using the Ins, Ctrl-Ins and Del keys on the keyboard

	keyboard.
ocRect3D	Rectangle nodes (Shape = shRectangle) are displayed with the three-dimensional effect. For rectangle nodes only.
ocAnimate	Expands and collapses nodes with an animation effect.

Rotated property**read/write**

Rotated controls the orientation of the tree structure layout.

property Rotated: Boolean;

Description

Use Rotated to flip the component's display 90 degrees. If Rotated=True, nodes of the same level are positioned vertically and parents are on the left in relation to child nodes.

Set Rotated to False for returning the component to its normal state with a horizontal layout of child nodes.

Selected property**read/write**

Selected contains a reference to the currently selected node in a tree.

property Selected: TdxOcNode;

Description

Read Selected to access a selected node. Within Options, if ocSelect is set to True, a focus rectangle highlights the selected node. If Selected=nil, then the component will not include a selected node. Setting a new value cancels the selection of the previous node and instead selects the pointed node. Once a node is selected, the OnChanging and OnChanged events are called.

See Also:

*TdxCustomOrgChart.DoChange, TdxCustomOrgChart.DoChanging,
TdxCustomOrgChart.OnChange, TdxCustomOrgChart.OnChanging*

TopEdge property**read/write**

TopEdge is the position of a window's upper edge relative to the component's upper edge.

property TopEdge: Integer;

Description

Read TopEdge to determine which area of the component is visible. Set TopEdge to change this visible area. When TopEdge changes, the window scrolls vertically.

See Also:

TdxCustomOrgChart.LeftEdge, TdxCustomOrgChart.SetLeftTop

Zoom property**read/write**

Turns the component's zooming mode on or off.

property Zoom: Boolean;

Description

Use Zoom to display the entire tree structure within the component window at runtime. When Zoom=True, the component diminishes to the window size (when necessary) and never displays

scroll bars. In this mode, a user cannot edit a node's text. Set Zoom to False for returning the component to its normal size.

Note

Value of the Zoom property does not reflect the real zooming state, because even if Zoom is True, the component size (client area) may be enough so as not to perform zooming. To know whether Zoom is effect, you should use the IsZoomed function instead of the Zoom property.

See Also:

TdxCustomOrgChart.DoZoom, TdxCustomOrgChart.IsZoomed

TdxCustomOrgChart Methods

Add	public
AddChild	public
AddChildFirst	public
AddFirst	public
AssignData	public
BeginUpdate	public
Clear	public
Create	public
CreateEditor	protected
CreateNode	protected
CreateParams	protected
DbClick	protected
Delete	public
Destroy	public
DoChange	protected
DoChanging	protected
DoZoom	public
DragDrop	public
DragOver	protected
DrawNode	protected
EndUpdate	public
FullCollapse	public
FullExpand	public
FullHeight	public
FullWidth	public
GetFirstNode	public
GetHitTestsAt	public
GetNodeAt	public
HideEditor	public
Insert	public
InternalAdd	protected
InternalMoveTo	protected
Invalidate	public
IsZoomed	public
KeyDown	protected

KeyPress	protected
Loaded	protected
MouseDown	protected
MouseMove	protected
NodeChanged	protected
Paint	protected
SetLeftTop	public
ShowEditor	public

Add method

Adds a new node to the tree structure.

function Add(Node: TdxOcNode; Ptr: Pointer): TdxOcNode;

Description

Creates a new node and adds it to the same level as the Node - to the end of the list. The Ptr parameter sets the Data property value of a new node. A reference to a new node returns as a result of the function call.

See Also:

TdxCustomOrgChart.AddChild, TdxCustomOrgChart.AddChildFirst, TdxCustomOrgChart.AddFirst, TdxCustomOrgChart.Insert

AddChild method

Adds a new node to the tree structure.

function Add(Node: TdxOcNode; Ptr: Pointer): TdxOcNode;

Description

Creates a new node and makes it the last child of the Node. The Ptr parameter sets the Data property value of a new node. A reference to a new node returns as a result of the function call.

See Also:

TdxCustomOrgChart.Add, TdxCustomOrgChart.AddChildFirst, TdxCustomOrgChart.AddFirst, TdxCustomOrgChart.Insert

AddChildFirst method

Adds a new node to the tree structure.

function AddChildFirst(Node: TdxOcNode; Ptr: Pointer): TdxOcNode;

Description

Creates a new node and makes it the first child of the Node. The Ptr parameter sets the Data property value of a new node. A reference to a new node returns as a result of calling the function.

See Also:

TdxCustomOrgChart.Add, TdxCustomOrgChart.AddChild, TdxCustomOrgChart.AddFirst, TdxCustomOrgChart.Insert

AddFirst method

Adds a new node to the tree structure.

```
function AddFirst(Node: TdxOcNode; Ptr: Pointer): TdxOcNode;
```

Description

Creates a new node and adds it to the same level as the Node - to the beginning of the list. The Ptr parameter sets the Data property value of a new node. A reference to the new node returns as a result of calling the function.

See Also:

TdxCustomOrgChart.Add, *TdxCustomOrgChart.AddChild*, *TdxCustomOrgChart.AddChildFirst*, *TdxCustomOrgChart.Insert*

AssignData method

Copies the data structure from another component.

```
procedure AssignData(Source: TdxCustomOrgChart);
```

Description

Copies data (i.e tree of nodes) from another TdxCustomOrgChart specified by the Source parameter. In this operation, previous data is lost.

BeginUpdate method

Prevents component redrawing until the EndUpdate method is called.

```
procedure BeginUpdate;
```

Description

Prohibits any changes to the screen until an EndUpdate call. With a large amount of additions and deletions, an application can call BeginUpdate, make some changes and then call EndUpdate for component redraw. The calls of BeginUpdate are accumulated, that's why each BeginUpdate call must have a corresponding EndUpdate call.

See Also:

TdxCustomOrgChart.EndUpdate

Clear method

Deletes all tree nodes managed by a component.

```
procedure Clear;
```

Description

Use the Clear method to remove all nodes from a tree structure.

See Also:

TdxCustomOrgChart.Delete, *TdxOcNode.DeleteChildren*, *TdxOcNode.HasChildren*

Create method

Creates an instance of TdxCustomOrgChart.

```
constructor Create(AOwner: TComponent); override;
```

Description

Creates the TdxCustomOrgChart component. The following values are assigned once a component is created.

Width	320
Height	200
DefaultNodeWidth	64
DefaultNodeHeight	40
IndentX	16
IndentY	16
LineWidth	1
BorderStyle	bsSingle
EditMode	[emWrap]
Options	[ocSelect,ocFocus,ocButtons,ocDbClick, ocEdit,ocCanDrag,ocShowDrag]

CreateEditor method

```
function CreateEditor: TdxOcInplaceEdit; virtual;
```

Description

Called when first entering text editing mode. Creates the TdxOcInplaceEdit object and returns a reference to it. Can be overridden to set some properties of the text editor or to create another editor inherited from the TdxOcInplaceEdit.

```
TMyChart = class(TdxCustomOrgChart)
    ...
    function CreateEditor: TdxOcInplaceEdit; override;
    ...
end;

function TMyChart.CreateEditor: TdxOcInplaceEdit;
begin
    Result := TdxOcInplaceEdit.Create(Self);
    Result.MaxLength := 40;      // editor with text length restriction
end;
```

See Also:

TdxCustomOrgChart.Editing, TdxCustomOrgChart.HideEditor, TdxCustomOrgChart.ShowEditor

CreateNode method

CreateNode creates a new tree node instance.

```
function CreateNode: TdxOcNode; virtual;
```

Description

Called from the InternalAdd, to create a new node. By default creates a TdxOcNode object, calls the OnCreateNode event handler and returns a reference to the created object. Descendants of the TdxCustomOrgChart can override the CreateNode for the creation of objects inherited from the

TdxOcNode. The overridden CreateNode method must create an object of the demanded class and as a result return a reference to it.

The created node is not a part of the data structure yet, that's why any operations are not available for it, except reading and setting properties. Do not call the CreateNode directly; use Add or Insert to create new nodes.

```
TMyNode = class(TdxOcNode)
    ...
end;

TMyChart = class(TdxCustomOrgChart)
    ...
    function CreateNode: TdxOcNode; override;
    ...
end;

function TMyChart.CreateNode: TdxOcNode;
begin
    Result := TMyNode.Create(Self); // a node of TmyNode class
    Result.Color := clRed;          // all the nodes are red
end;
```

See Also:

TdxCustomOrgChart.OnCreateNode.

CreateParams method

CreateParams initializes parameters for window creation.

```
procedure CreateParams(var Params: TCreateParams); override;
```

Description

Overrides the TWinControl.CreateParams. Includes scroll bars into the window and sets a border style according to the BorderStyle and Ctl3D properties.

See Also:

TdxCustomOrgChart.BorderStyle

DbClick method

DbClick processes the mouse button double click event.

```
procedure DbClick; override;
```

Description

Overrides the TWinControl.DbClick. If the ocDbClick in Options is True, then changes pointed by the mouse node's Expanded property to the opposite, i.e. expands or collapses the node.

See Also:

TdxCustomOrgChart.MouseDown, TdxCustomOrgChart.Options

Delete method

Removes a node from the tree structure.

```
procedure Delete(Node: TdxOcNode); virtual;
```

Description

Deletes a Node and all its children. Components inherited from TdxCustomOrgChart can override Delete for setting additional actions on deleting the node. The overridden method Delete must always end with calling the inherited Delete.

On overriding Delete you should take into account the following: first – the possibility of Delete (nil) call and second – the possibility of recursive calls of Delete via DeleteChildren, but in this case references to some nodes may be invalid. If the overridden method Delete references other nodes, then you should check the Deleting property as in the example below.

```
procedure TMyChart.Delete(Node: TdxOcNode);
var
  N: TdxOcNode;
begin
  if Node = nil then Exit;          // obligatory!
  Images.Delete(Node.ImageIndex);   // (for example)
  N := GetFirstNode;
  while N <> nil do begin           // a cycle in all nodes
    if N.Deleting then N := Node; // ignores already deleted nodes!
    if N.ImageIndex > Node.ImageIndex then
      N.ImageIndex := N.ImageIndex - 1;
    N := N.GetNext;
  end;
  inherited Delete(Node);
end;
N.Deleting = True means, that N is a direct or indirect parent of the Node and all its children
foregoing the Node are already deleted.
```

See Also:

TdxOcNode.DeleteChildren, TdxOcNode.Deleting, TdxOcNode.HasChildren

Destroy method

Destroys an instance of TdxCustomOrgChart.

destructor Destroy; override;

Description

Destroys the TdxCustomOrgChart component...Before the inherited Destroy call destroys all tree nodes. Do not call Destroy directly; call Free instead.

DoChange method

DoChange notifies that a selected node has changed.

procedure DoChange(Node: TdxOcNode); virtual;

Description

Called after a Node is selected. By default calls the OnChange event, if it is defined. Can be overridden to provide reaction to the change of a selected node.

See Also:

TdxCustomOrgChart.DoChanging, TdxCustomOrgChart.OnChange, TdxCustomOrgChart.Selected

DoChanging method

procedure DoChanging(Node: TdxOcNode; var Allow: Boolean); virtual;

Description

Called before a Node is selected. By default calls the OnChanging event, if it is defined. The overridden method must set Allow to True or False based on whether the selection of a Node is allowed or prohibited. DoChanging is called on selection of a node by the mouse or keys, but is not called on the program setting of the Selected property.

See Also:

TdxCustomOrgChart.DoChange, TdxCustomOrgChart.OnChanging, TdxCustomOrgChart.Selected

DoZoom method

Scales the given value in Zoom mode.

function DoZoom(Value: Integer): Integer;

Description

If Zoom is False, it returns the unchanged value of the Value parameter. If Zoom is True, returns Value diminished in accordance with a scaling factor. DoZoom can be used by applications for font scaling in the OnSetFont event, for example.

See Also:

TdxCustomOrgChart.IsZoomed, TdxCustomOrgChart.Zoom

DragDrop method

Completes a Drag & Drop operation.

procedure DragDrop(Source: TObject; X, Y: Integer); override;

Description

Overrides TControl.DragDrop. If the OnDragDrop event is defined then calls it; Otherwise having called MoveTo, moves the selected node to that pointed by the mouse position. This scheme works for inner movements within one TdxOrgChart component only. For Drag & Drop implementation between and among multiple components, the OnDragDrop must be defined. OnDragOver event is not required if a source of movement belongs to TdxCustomOrgChart class.

See Also:

TdxCustomOrgChart.DragOver

DragOver method

DragOver processes dragging of an object over the component.

procedure DragOver(Source: TObject; X, Y: Integer; State: TDragState;
var Accept: Boolean); override;

Description

Overrides the TControl.DragOver. Enables or disables Drag and Drop operations depending on cursor location. Scrolls the window once it approaches its edges. If the ocShowDrag in Options is True, draws the dragged node's contour and its connections with the potential new parent. Finally calls the OnDragOver event, if it is defined.

See Also:

TdxCustomOrgChart.DragDrop, TdxCustomOrgChart.Options

DrawNode method

DrawNode draws a tree node on the specified canvas.

```
procedure DrawNode(Node: TdxOcNode; ACanvas: TCanvas; Rect: TRect);  
virtual;
```

Description

Called on drawing a Node on the screen. By default calls the OnDfawNode event. If undefined, displays the text and the image defined by the Node.Text and Node.ImageIndex properties. Overriding the DrawNode, the TdxCustomOrgChart descendants can implement another algorithm of node drawing. Do not use the Canvas property of the component for drawing; use ACanvas parameter instead. Rect parameter contains the coordinates of a Node's client area, which must limit the drawing region. The DfawNode method must not change the properties of the nodes, otherwise endless recursion of component drawing is possible.

See Also:

TdxCustomOrgChart.OnDrawNode

EndUpdate method

Re-enables screen repainting after the BeginUpdate call.

```
procedure EndUpdate;
```

Description

Cancels BeginUpdate and calls Invalidate for component redrawing. After this, any changes will be displayed on the screen. The calls of BeginUpdate are accumulated, that's why each BeginUpdate call must have a corresponding EndUpdate call.

See Also:

TdxCustomOrgChart.BeginUpdate

FullCollapse method

Collapses all nodes in the OrgChart.

```
procedure FullCollapse;
```

Description

Call FullCollapse to hide all nodes of the tree, leaving visible only the nodes of the top level.

See Also:

TdxCustomOrgChart.FullExpand, TdxOcNode.Collapse

FullExpand method

Expands all nodes in the OrgChart.

```
procedure FullExpand;
```

Description

Call FullExpand to display all nodes of the tree, making a data structure completely visible on the screen. If ocButtons in Options is True, all node buttons will change from {+} to {-}.

See Also:

TdxCustomOrgChart.FullCollapse, TdxOcNode.Expand

FullHeight method

Returns the current OrgChart height.

```
function FullHeight: Integer;
```

Description

Call FullHeight to examine the full height of a component, which may be bigger than the window height. This value changes upon additions to and deletions from nodes, expanding and collapse, as well as on zooming. Unlike the TdxOcNode.Height property, FullHeight always returns the actual height, regardless of the Rotated property.

See Also:

TdxCustomOrgChart.FullWidth, TdxOcNode.FullHeight

FullWidth method

Returns the current OrgChart width.

```
function FullWidth: Integer;
```

Description

Call FullWidth to examine the full width of a component, which may be bigger than the window width. This value changes upon additions and deletions from nodes, expanding and collapse, as well as zooming. Unlike the TdxOcNode.Width property, FullWidth always returns the actual Width, regardless of the Rotated property.

See Also:

TdxCustomOrgChart.FullHeight, TdxOcNode.FullWidth

GetFirstNode method

Returns the first node in a tree.

```
function GetFirstNode: TdxOcNode;
```

Description

Use GetFirstNode to retrieve the first node of a tree, i.e. the very left node of the zero level. GetFirstNode is similar to Items[0] but returns nil if there are no nodes.

GetHitTestsAt method

Returns information about the point location in a component's client area.

Type:

```
TdxOcHitTest = (htNowhere, htOnLeftIndentX, htOnRightIndentX, htOnIndentY,
htUnder, htOnRect, htOnShape, htOnButton);
TdxOcHitTests = set of TdxOcHitTest;
```

```
function GetHitTestsAt(X,Y: Integer): TdxOcHitTests;
```

Description

Returns flag combination, determining the location of a point with (X, Y) coordinates relative to the node returned by GetNodeAt function. Coordinates are set relative to the client area of the component...Usually used while processing the events of a mouse. Values of the flags are the following:

Value	Location of (X, Y)
htNowhere	Does not belong to any node.
htOnLeftIndentX	In the separating space area to the left of the node. (Rotated=True above a node)

	above a node).
htOnRightIndentX	In the separating space area to the right of the node. (Rotated=True – below a node).
htOnIndentY	In the separating area above the node. (Rotated=True – to the left of the node).
htUnder	Below a closed node or a node without children (Rotated=True- to the right of a node).
htOnRect	Inside a rectangle, limiting the node's contour.
htOnShape	Inside the node's contour. For rectangle nodes htOnShape and htOnRect are similar.
htOnButton	On the [+] / [-] button.

See Also:

TdxCustomOrgChart.GetNodeAt

GetNodeAt method

Returns the node that is found at a specified position.

function GetNodeAt(X,Y: Integer): TdxOcNode;

Description

Returns a reference to the node, containing a point with (X, Y) coordinates. Coordinates are set relative to the client area of the component...Usually used in handling mouse events. Use the GetHitTestsAt function to determine the point location relative to this node. If no node is found, GetNodeAt returns nil.

The example below offers an implementation of a simple Drag & Drop interface with an ability of node shifts between components.

```

procedure TForm1.Chart1DragDrop(Sender, Source: TObject; X,Y: Integer);
var
  Node, New: TdxOcNode;
  Mode: TdxOcAttachMode;
  HT: TdxOcHitTests;
  Chart: TdxOrgChart;
begin
  Chart := TdxOrgChart(Sender);           // component-receiver
  Node := Chart.GetNodeAt(X,Y);           // indicated by the mouse node
  HT := Chart.GetHitTestsAt(X,Y);         // or its area
  Mode := naInsert;
  if htUnder in HT then Mode := naAddChild;
  if HT * [htOnRect,htOnRightIndentX] <> []
  then Node := Node.GetNextSibling;       // (insertion after the Node)
  if Node=nil then begin                  // (if the Node is the last)
    Node := Chart.GetNodeAt(X,Y).Parent;
    Mode := naAddChild;
  end;
  if Source=Chart
  then Chart.Selected.MoveTo(Node,Mode) // internal drag
  else begin                             // external drag
    if Mode=naInsert then New := Chart.Insert(Node,nil)
    else New := Chart.AddChild(Node,nil);
    Node := TdxOrgChart(Source).Selected;
    New.Text := Node.Text;
  end;
end;

```

```
New.ImageIndex := Node.ImageIndex; //(Sender and Source have one
and the same Image list)
New.ImageAlign := Node.ImageAlign; //
Chart.Selected := New;
end;
Chart.Selected.MakeExposed;
end;
```

See Also:

TdxCustomOrgChart.GetHitTestsAt

HideEditor method

Ends node text editing.

```
procedure HideEditor(Save: Boolean);
```

Description

If edit mode is active, finishes the text editing of a node. The Save parameter indicates whether to save the entered text. If Save=True, the Text property of the selected node will be set to the entered text; if Save=False, the node's Text property stays unchanged. Setting the Editing property to False is similar to HideEditor(True).

See Also:

TdxCustomOrgChart.Editing, TdxCustomOrgChart.ShowEditor

Insert method

Inserts a new node to a tree structure.

```
function Insert(Node: TdxOcNode; Ptr: Pointer): TdxOcNode;
```

Description

Creates a new node and adds it to the data structure directly before the Node parameter. The Ptr parameter sets the Data property value of a new node. A reference to the new node returns as a result of the function call.

See Also:

TdxCustomOrgChart.Add, TdxCustomOrgChart.AddChild, TdxCustomOrgChart.AddChildFirst, TdxCustomOrgChart.AddFirst

InternalAdd method

Adds a new node to the tree data structure.

```
procedure InternalAdd(AParent: TdxOcNode; Ptr: Pointer; AIndex: Integer):
TdxOcNode; virtual;
```

Description

Called from the Add... and Insert methods. Creates a new node by calling CreateNode and inserts it in the AParent child list by AIndex. The Data property is set to Ptr. If AParent =nil, a new node is placed at the top level of the hierarchy. A reference to the created node returns as a result of a function call. TdxCustomOrgChart descendants can override InternalAdd for providing additional actions on creating new nodes. The overridden InternalAdd method must always call inherited InternalAdd. A direct call of the InternalAdd should be used carefully, since incorrect settings of parameters may raise an exception.

See Also:

TdxCustomOrgChart.Add, TdxCustomOrgChart.AddChild, TdxCustomOrgChart.AddChildFirst, TdxCustomOrgChart.AddFirst, TdxCustomOrgChart.CreateNode, TdxCustomOrgChart.Insert

InternalMoveTo method

Moves a node to a new location in the tree structure.

```
procedure InternalMoveTo(AParent, ANode: TdxOcNode; AIndex: Integer);
virtual;
```

Description

Called from MoveTo. Moves the ANode to the AParent children list by AIndex. If AParent=nil, the node moves to the top level of the hierarchy. TdxCustomOrgChart descendants can override InternalMoveTo to provide additional actions when moving nodes. The overridden InternalMoveTo method must always call inherited InternalMoveTo. A direct call of the InternalMoveTo should be used carefully since incorrect settings of the parameter may raise exceptions.

See Also:

TdxOcNode.MoveTo

Invalidate method

Notifies that a component must be repainted.

```
procedure Invalidate; override;
```

Description

Overrides TControl.Invalidate. Calls inherited Invalidate only when necessary.

IsZoomed method

Determines whether zooming is active.

```
function IsZoomed: Boolean;
```

Description

Returns True if Zoom mode is both True and Active, i.e. the linear size of the component is diminished to the window size.

See Also:

TdxCustomOrgChart.Zoom, TdxCustomOrgChart.DoZoom

KeyDown method

Processes keyboard input.

```
procedure KeyDown(var Key: Word; Shift: TShiftState); override;
```

Description

Overrides the TWinControl.KeyDown. Implements cursor key functionality. If ocInsDel within Options is set to True, it also implements the following key functions: Ins (inserts a new node), Ctrl-Ins (inserts a new child) and Del (deletes a node).

See Also:

TdxCustomOrgChart.KeyPress

KeyPress method

Processes signals from a keyboard's alphanumeric keys.

procedure KeyPress(var Key: Char); override;

Description

Overrides the TWinControl.KeyPress. Processes the {+} and {} keys, used for expanding and collapsing a selected node.

See Also:

TdxCustomOrgChart.KeyDown

Loaded method

Ends the process of component loading from the stream.

procedure Loaded; override;

Description

Overrides the TComponent.Loaded. At design-time, upon loading of the component, calls FullExpand.

MouseDown method

Processes signals from mouse buttons.

procedure MouseDown(Button: TMouseButton; Shift: TShiftState; X,Y: Integer);
override;

Description

Overrides the TControl.MouseDown. Clicking a node via the left mouse button selects a given node. Clicking the {+} and {} buttons within the chart, expands or collapses a given node.

See Also:

TdxCustomOrgChart.MouseMove

MouseMove method

Processes mouse movement over the component.

procedure MouseMove(Shift: TShiftState; X,Y: Integer); override;

Description

Overrides the TControl.MouseMove. If the left button is pressed and the ocCanDrag mode in Options is set to True, MouseMove starts the Drag & Drop process.

See Also:

TdxCustomOrgChart.MouseDown, TdxCustomOrgChart.Options

NodeChanged method

Notifies that one or more node properties have changed.

procedure NodeChanged(Node: TdxOcNode); virtual;

Description

Called on changes made to any of the following properties of a Node: Width, Height, ChildAlign, Color, Shape, ImageIndex, ImageAlign or Text. By default, this element performs no actions. TdxCustomOrgChart descendants can override this method in order to provide reaction to changes of node properties.

Paint method

Draws a component on the screen.

procedure Paint; override;

Description

Overrides the TCustomControl.Paint. Draws the component by the system message WM_PAINT.

SetLeftTop method

Sets a new component location in its window.

procedure SetLeftTop(ALeft,ATop: Integer);

Description

Use SetLeftTop to set new values for the LeftEdge and TopEdge properties. This operation scrolls the window in one or both directions.

Note: The set values of the LeftEdge and TopEdge properties might differ from ALeft and ATop parameters due to scrolling restrictions.

See Also:

TdxCustomOrgChart.LeftEdge, TdxCustomOrgChart.TopEdge

ShowEditor method

Begins text editing of a node.

procedure ShowEditor; virtual;

Description

Use ShowEditor to start text editing of a selected node. ShowEditor is ignored if the ocEdit mode in Options is False or there is no selected item or the component is in the Zoom mode. The ShowEditor call is equal to setting Editing = True.

See Also:

TdxCustomOrgChart.Editing, TdxCustomOrgChart.HideEditor

TdxCustomOrgChart Events

All events are protected.

OnChange	OnChanging	OnCollapsed
OnCollapsing	OnCreateNode	OnDeletion
OnDrawNode	OnEdited	OnEditing
OnExpanded	OnExpansion	OnGetText
OnSetFont	OnSetText	

OnChange event

Occurs when the selection has changed from one node to another.

TdxOcEvent = **procedure**(Sender: TObject; Node: TdxOcNode) of object;
property OnChange: TdxOcEvent;

Description

Called after a Node is selected (Selected). On changing a selected node by the OnChange event can fulfill definite actions. The example below is taken from the node editor for TdxOrgChart at design-time. On selection of any node, its properties are displayed in the window and become available for editing.

```
procedure TEditor.ChartChange(Sender: TObject; Node: TdxOcNode);
var
  Info: TdxOcNodeInfo;
begin
  Node.GetNodeInfo(Info);
  WidthEdit.Text := IntToStr(Info.Width);    // (TEdit)
  HeightEdit.Text := IntToStr(Info.Height);  // (TEdit)
  ImIndexEdit.Text := IntToStr(Info.Index);  // (TEdit)
  ImAlignEdit.ItemIndex := Ord(Info.IAlign); // (TComboBox)
  ChAlignEdit.ItemIndex := Ord(Info.Align);  // (TComboBox)
  ShapeEdit.ItemIndex := Ord(Info.Shape);    // (TComboBox)
  ColorEdit.Text := ColorToString(Info.Color); // (TComboBox)
end;
```

See Also:

TdxCustomOrgChart.DoChange, TdxCustomOrgChart.OnChanging, TdxCustomOrgChart.Selected

OnChanging event

Occurs whenever a selection is about to change from one node to another.

TdxOcAllowEvent = **procedure**(Sender: TObject; Node: TdxOcNode;
 var Allow: Boolean) of object;
property OnChanging: TdxOcAllowEvent;

Description

Called before a Node is selected (Selected). The processing procedure must set Allow to True or False depending on whether the Node selection is enabled or disabled. The OnChanging property is called upon selection of a node by the mouse or keys, but not called by the program setting of the Selected property.

```

procedure TForm1.ChartChanging(Sender: TObject; Node: TdxOcNode;
  var Allow: Boolean);
begin
  Allow := Node.Color <> clRed; // not to select the red nodes
end;

```

See Also:

TdxCustomOrgChart.DoChange, TdxCustomOrgChart.OnChanging, TdxCustomOrgChart.Selected

OnCollapsed event

TdxOcEvent = **procedure**(Sender: TObject; Node: TdxOcNode) of object;
property OnCollapsed: TdxOcEvent;

Description

Called after the collapsing the Node by Collapse method. Use the OnCollapsed event to fulfill definite actions while closing the nodes.

See Also:

TdxCustomOrgChart.OnCollapsing, TdxOcNode.Collapse

OnCollapsing event

Occurs after a node has collapsed.

TdxOcAllowEvent = **procedure**(Sender: TObject; Node: TdxOcNode;
 var Allow: Boolean) of object;
property OnCollapsing: TdxOcAllowEvent;

Description

Called after collapsing a Node by the Collapse method. Use the OnCollapsed event to perform actions after collapsing nodes.

See Also:

TdxCustomOrgChart.OnCollapsed, TdxOcNode.Collapse

OnCreateNode event

Occurs when a new node has been created.

TdxOcCreateEvent = **procedure**(Sender: TObject; var Node: TdxOcNode) of object;
property OnCreateNode: TdxOcCreateEvent;

Description

Called on creating a new node by Add... and Insert methods. Having called Create the processing procedure must create a new node and assign its reference to the Node. It's possible to set definite properties of a new node or create a new object class inherited from the TdxOcNode. Created in OnCreateNode procedure node is not a part of the data structure yet, that's why only read and set properties are available.

```

procedure TForm1.ChartCreateNode(Sender: TObject; var Node: TdxOcNode);
begin
  Node := TdxOcNode.Create;           // node creation
  Node.Shape := shEllipse; // all nodes are in the form of an ellipse
  Node.Color := clYellow; //yellow color
end;

```

See Also:

TdxCustomOrgChart.CreateNode, TdxOcNode.Create

OnDeletion event

Occurs when a node in the OrgChart is deleted.

```
TdxOcEvent = procedure(Sender: TObject; Node: TdxOcNode) of object;  
property OnDeletion: TdxOcEvent;
```

Description

Called on deleting a Node by the Delete method. Use the OnDeletion event to perform actions when d

See Also:

TdxCustomOrgChart.DrawNode

OnEdited event

Occurs after a user edits the Text property of a node.

```
TdxOcGetTextEvent = procedure(Sender: TObject; Node: TdxOcNode;  
    var Value: String) of object;  
property OnEdited: TdxOcGetTextEvent;
```

Description

Called on ending text editing for a Node. The Value parameter contains the text entered into the editor.

Before the Node.Text property assigns this value, the processing procedure can alter it. If editing has been canceled by the Esc key or HideEditor(False) call, OnEdited is not called.

See Also:

TdxCustomOrgChart.HideEditor, TdxCustomOrgChart.OnEditing

OnEditing event

Occurs when a user starts to edit the Text property of a node.

```
TdxOcAllowEvent = procedure(Sender: TObject; Node: TdxOcNode;  
    var Allow: Boolean) of object;  
property OnEditing: TdxOcAllowEvent;
```

Description

Called before the call of the text editor mode for a Node. The processing procedure can set Allow to False in order to disable editing.

See Also:

TdxCustomOrgChart.OnEdited

OnExpanded event

Occurs after expanding a node.

```
TdxOcEvent = procedure(Sender: TObject; Node: TdxOcNode) of object;  
property OnExpanded: TdxOcEvent;
```

Description

Called after the expansion of a Node by Expand method. Use the OnExpand event to perform actions when expanding nodes.

See Also:

TdxCustomOrgChart.OnExpansion, TdxOcNode.Expand

OnExpansion event

Occurs after expanding a node.

```
TdxOcAllowEvent = procedure(Sender: TObject; Node: TdxOcNode;
    var Allow: Boolean) of object;
property OnExpansion: TdxOcAllowEvent;
```

Description

Called before the expansion the a Node by Expand method. The processing procedure must set Allow to True or False depending on whether Node expansion is enabled or disabled.

C

See Also:

TdxCustomOrgChart.OnExpand, TdxOcNode.Expand

OnGetText event

Occurs on reading the text of a node.

```
TdxOcGetTextEvent = procedure(Sender: TObject; Node: TdxOcNode;
    var Value: String) of object;
property OnGetText: TdxOcGetTextEvent;
```

Description

Implements reading the Node.Text property. The processing procedure must return a text string via Value. Using the OnGetText event, an application can implement its own unique requirements that differ from the standard access scheme to texts: for example, to create a node's number by its Level and Index properties.

```
procedure TForm1.ChartGetText(Sender: TObject; Node: TdxOcNode;
    var Value: String);
begin
    Value := IntToStr(Node.Index+1);
    if Node.Parent<>nil then Value := Node.Parent.Text + '.' + Value;
end;
```

See Also:

TdxCustomOrgChart.OnSetText, TdxOcNode.GetText

OnSetFont event

Occurs when a node requires drawing its text.

```
TdxOcFontEvent = procedure(Sender: TObject; Node: TdxOcNode; Font: TFont)
    of object;
property OnSetFont: TdxOcFontEvent;
```

Description

Called on drawing a Node on the screen and also on calling the text editor for the Node. The processing procedure can alter any Font properties, i.e. add individual fonts for any concrete node.

See Also:

TdxOcNode.SetFont

OnSetText event

Occurs on setting the text of a node.

```
TdxOcSetTextEvent = procedure(Sender: TObject; Node: TdxOcNode;  
const Value: String) of object;  
property OnSetText: TdxOcSetTextEvent;
```

Description

Saves the Node.Text property. Using the OnSetText event, an application can implement its own unique requirements that differ from the standard access scheme to text: for example, store text strings in a TStringList list or in a text file.

See Also:

TdxCustomOrgChart.OnGetText, TdxOcNode.SetText

TdxOrgChart

Unit

DxOrgChr

Description

TdxOrgChart – is a fully functional component designed for the display and editing of hierarchical tree data structures such as the contents of a book or file system on a disc. Each node of the tree can have a free number of children, displayed horizontally, directly below a given node. TdxOrgChart enables an incredibly wide number of possibilities to manage the contents and appearance of both the entire chart structure as well as each individual node.

TdxOrgChart Properties

Derived from TdxCustomOrgChart

BorderStyle	published/read/write
Count	public/read/write
DefaultImageAlign	published/read/write
DefaultNodeHeight	published/read/write
DefaultNodeWidth	published/read/write
Editing	public/read/write
EditMode	published/read/write
Images	published/read/write
IndentX	published/read/write
IndentY	published/read/write
Items	public/read/write
LeftEdge	public/read/write
LineWidth	published/read/write
Options	published/read/write
Rotated	published/read/write
Selected	public/read/write
TopEdge	public/read/write
Zoom	public/read/write

TdxOrgChart Methods

In TdxOrgChart

DefineProperties public

Derived from TdxCustomOrgChart

Add	AddChild	AddChildFirst	AddFirst
AssignData	BeginUpdate	Clear	Create
CreateEditor	CreateNode	CreateParams	DblClick
Delete	Destroy	DoChange	DoChanging
DoZoom	DragDrop	DragOver	DrawNode
EndUpdate	FullCollapse	FullExpand	FullHeight
FullWidth	GetFirstNode	GetHitTestsAt	GetNodeAt
HideEditor	Insert	InternalAdd	InternalMoveTo
Invalidate	IsZoomed	KeyDown	KeyPress
Loaded	MouseDown	MouseMove	NodeChanged
Paint	SetLeftTop	ShowEditor	

DefineProperties method

```
procedure DefineProperties(Filer: TFiler); override;
```

Description

Overrides TWinControl.DefineProperties. After the saving published properties stores the node structure connected with the component.

TdxOrgChart Events

Derived from TdxCustomOrgChart

OnCreateNode	OnChange	OnChanging	OnCollapsed
OnCollapsing	OnDeletion	OnExpanded	OnExpansion
OnEditing	OnEdited	OnSetFont	OnDrawNode
OnGetText	OnSetText		

TdxOcInplaceEdit

Unit

DxOrgChr

Description

TdxOcInplaceEdit is a multiline text editor for text editing of TdxCustomOrgChart nodes. Created and controlled by the TdxCustomOrgChart component, it is not intended for use outside it.

TdxOcInplaceEdit Methods

CreateParams	protected
KeyDown	protected
KeyUp	protected
KeyPress	protected
Create	public
Tree	public

CreateParams method

```
procedure CreateParams(var Params: TCreateParams); override;
```

Description

Overrides TCustomEdit.CreateParams. Sets the edit modes according to the set ones in Tree.EditMode.

See Also:

TdxOcInplaceEdit.Create

KeyDown method

```
procedure KeyDown(var Key: Word; Shift: TShiftState); override;
```

Description

Overrides TCustomEdit.KeyDown. Handles the following keys:

F2, Alt-Enter – ends editing;

Esc – cancels editing;

Alt + arrows – stops editing coming to the next node.

See Also:

TdxOcInplaceEdit.KeyPress, TdxOcInplaceEdit.KeyUp

KeyUp method

```
procedure KeyUp(var Key: Word; Shift: TShiftState); override;
```

Description

Overrides TCustomEdit.KeyUp. Calls the KeyUp method of its owner.

See Also:

TdxOcInplaceEdit.KeyPress, TdxOcInplaceEdit.KeyDown

KeyPress method

procedure KeyPress(var Key: Char); override;

Description

Overrides TCustomEdit.KeyPress. First calls KeyPress of its owner, then if a key is not filtered, calls inherited KeyPress and in case of necessity alters its own size.

See Also:

TdxOcInplaceEdit.KeyUp, TdxOcInplaceEdit.KeyDown

Create method

constructor Create(AOwner: TComponent); override;

Description

Creates the TdxOcInplaceEdit object. After calling inherited Create sets Visible and Ctl3D properties to False and Parent equal to AOwner.

See Also:

TdxOcInplaceEdit.CreateParams

Tree method

function Tree: TdxCustomOrgChart;

Description

Returns reference to the TdxCustomOrgChart component having the given TdxOcInplaceEdit object.

TdxOcNode

Unit

DxOrgChr

Description

TdxOcNode – is a class, describing a single node in the TdxCustomOrgChart tree data structure. Each node can have its own size, shape, color, text and image.

TdxOcNode Properties

ChildAlign	public/read/write
Color	public/read/write
Count	public/read/write
Data	public/read/write
Deleting	public/read/write
Expanded	public/read/write
HasChildren	public/read/write
Height	public/read/write
ImageAlign	public/read/write
ImageIndex	public/read/write
Index	public/read/write
IsVisible	public/read/write
Items	public/read/write
Level	public/read/write
Owner	public/read/write
Parent	public/read/write
Selected	public/read/write
Shape	public/read/write
Text	public/read/write
Width	public/read/write

ChildAlign property

read/write

Determines a node's position in its home area.

```
TdxOcNodeAlign = (caLeft,caCenter,caRight)
property ChildAlign: TdxOcNodeAlign;
```

Description

Use ChildAlign to set the location of a node on the screen relative to its children. The following variants are possible:

caLeft

caCenter

caRight

Color property

read/write

Background color of a node.

```
property Color: TColor;
```

Description

Sets the color of the background for a given node. If Color = cNone, the color of a node is determined by the Color property of the TdxOrgChart component. Reading the Color property always returns an actual color instead of cNone.

Count property**read/write**

Then number of direct descendants of a node.

property Count: Integer;

Description

Contains the number of direct descendants of a given node. This number includes direct descendants only.

See Also:

TdxOcNode.Parent

Data property**read/write**

A pointer to application-defined data associated with a node.

property Data: Pointer;

Description

Contains a pointer to data determined by an application and connected with a given node. TdxOcNode and TdxCustomOrgChart methods do not use this property.

See Also:

TdxOcNode.SetData

Deleting property**read/write**

Indicates whether a node is in the process of being Deleted.

property Deleting: Boolean;

Description

The Deleted property is set to True while deleting a node and also (temporarily) while deleting its children by the DeleteChildren method. Used by Delete and Destroy methods in order to prevent an endless recursion on deleting nodes.

See Also:

TdxCustomOrgChart.Delete, TdxOcNode.DeleteChildren, TdxOcNode.Destroy

Expanded property**read/write**

Specifies whether a tree node is expanded.

property Expanded: Boolean;

Description

Expanded property is True, if a given node is expanded, i.e. its children are visible on the screen. Setting Expanded to True or False expands or collapses nodes accordingly. It does not always initiate changes on the screen, since a parent node can be collapsed or invisible. For nodes without children the Expanded property is always equal to False and cannot be set to True.

See Also:

TdxOcNode.Collapse, TdxOcNode.Expand

HasChildren property

read/write

Indicates whether a node has children list.

property HasChildren: Boolean;

Description

HasChildren is True, if a node has a list of children and False otherwise. Automatically set to True on adding children to the node. Setting HasChildren = False deletes all the nodes and the list itself.

Note: HasChildren property displays only the presence of the children list which might be also blank. Use the Count property to determine the actual presence of children.

See Also:

TdxOcNode.Count, TdxOcNode.DeleteChildren

Height property

read/write

A tree node's height in screen pixels.

property Height: Word;

Description

Use Height to set the height of a node on the screen. If Height=0, the height of a node is determined by the Owner.DefaultNodeHeight property. Reading the Height property always returns the actual height taking into account default and scaling. On setting Height, however, you should set either 0 or real (non-scaled) height.

See Also:

TdxCustomOrgChart.DefaultNodeHigh, TdxOcNode.Width

ImageAlign property

read/write

Determines image position for a node.

`TdxOcImageAlign = (iaNone, iaLT, iaLC, iaLB, iaRT, iaRC, iaRB, iaTL, iaTC, iaTR, iaBL, iaBC, iaBR);`

property ImageAlign: TdxOcImageAlign;

Description

Use ImageAlign to set the location of an image within a node. If ImageAlign=iaNone, the node does not have any image even when setting the ImageIndex property.

iaLT – Left Top

iaLC – Left Center

iaLB – Left Bottom

iaRT – Right Top

iaRC – Right Center

iaRB – Right Bottom

iaTL – Top Left

iaTC – Top Center

iaTR – Top Right

iaBL – Bottom Left

iaBC – Bottom Center

iaBR – Bottom Right

See Also:

TdxCustomOrgChart.Images, TdxOcNode.InageIndex

ImageIndex property

read/write

Specifies the image displayed within a tree node.

property ImageIndex: Smallint;

Description

Use ImageIndex to set the index of an image for a node in Owner.Images list.

If ImageIndex = -1, a node does not include an image. The image is visible when ImageAlign property is not equal to iaNone.

See Also:

TdxCustomOrgChart.Images, TdxOcNode.InageAlign

Index property

read/write

Index is the position of a node in the list of child nodes maintained by its parent node.

property Index: Integer;

Description

Use Index to determine the position of a node relative to its sibling nodes. Numbering begins with 0 from the left to the right, and subsequent children are indexed sequentially.

See Also:

TdxOcNode.IndexOf

IsVisible property

read/write

Indicates whether a node is currently visible in the tree structure image.

property IsVisible: Boolean;

Description

IsVisible is True, if a given node is visible on the screen, i.e. both its parent and all parents of the higher levels are expanded.

Note: IsVisible does not specify whether a node is in the window area or outside it.

See Also:

TdxOcNode.MakeVisible

Items property

read/write

Items is an indexed array of immediate children of a node.

property Items[Index:Integer]: TdxOcNode;

Description

Use Index to access a child of a given node by its index. This is a default property, so Node.Items[I] is equal to Node[I]. Items and Count properties are usually used to iterate all children of a node, for example:

```
for I := 0 to Count-1 do
    Items[I].Color := clRed;
```

See Also:

TdxOcNode.Count, TdxOcNode.Index

Level property

read/write

Indicates the total number of parents for the a node.

property Level: Integer;

Description

Use Level to determine a level number of a given node in the hierarchical data structure of a TdxCustomOrgChart object. Nodes of the top level with no parents are on level 0, their children are on level 1, etc.

Owner property

read/write

Owner is the TdxCustomOrgChart object having a tree node.

property Owner: TdxCustomOrgChart;

Description

Use Owner to refer to the TdxCustomOrgChart component which owns a given node.

Parent property

read/write

Identifies the parent node of a tree node.

property Parent: TdxOcNode;

Description

Use Parent to refer to the immediate parent of a given node. The Parent node is one level higher than a given node and includes the node as a child.

Selected property

read/write

Selected determines whether a node is selected.

property Selected: Boolean;

Description

Selected property is True, if a given node is selected in the tree structure. Set Selected to True or False to make a node selected or to cancel selection.

See Also:

TdxCustomOrgChart.Selected

Shape property

read/write

Shape determines the geometric image of the node frame.

TdxOcShape = (shRectangle, shRoundRect, shEllipse, shDiamond);

property Shape: TdxOcShape;

Description

Use Shape to set the form of a node for its display on the screen.

The following options are available:

shRectangle

shRoundRect

shEllipse

shDiamond

Text property**read/write**

Represents a text string associated with a node.

property Text: String;

Description

Use Text to set a text string connected with a given node and displayed on the screen.

See Also:

TdxOcNode.GetText, TdxOcNode.SetText

Width property**read/write**

Width of a tree node in screen pixels.

property Width: Word;

Description

Use Width to set the width of a node on the screen. If Width=0, the width of a node is determined by the Owner.DefaultNodeWidth property. Reading the Width property always returns the actual width taking into account default and scaling. On setting Width, however, you should set it to 0 or real (non-scaled) width.

TdxOcNode Methods

ChildrenHeight	public
ChildrenWidth	public
ClientRect	public
Collapse	public
Create	public
DeleteChildren	public
Destroy	public
DisplayRect	public
Expand	public
Focused	public
FullHeight	public
FullWidth	public
GetFirstChild	public
GetLastChild	public
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GetNextChild	public
GetNextSibling	public
GetNextVisible	public
GetNodeInfo	public
GetPrev	public
GetPrevChild	public
GetPrevSibling	public
GetPrevVisible	public
GetText	protected
HasAsParent	public
IndexOf	public
MakeExposed	public
MakeVisible	public
MoveTo	public
ReadData	protected
SetData	protected
SetFont	protected
SetText	protected
WriteData	protected

ChildrenHeight method

ChildrenHeight returns the height of a node's children.

```
function ChildrenHeight: Integer;
```

Description

Returns the total height of the area occupied by all children of a given node. The returned value depends on scaling. If Owner.Rotated=True, returns the total width of the children area instead of the height. If the node is collapsed (Expanded=False), then returns 0.

See Also:

TdxOcNode.ChildrenWidth, TdxOcNode.FullHeight

ChildrenWidth method

ChildrenWidth returns the width of a node's children.

```
function ChildrenWidth: Integer;
```

Description

Returns the total width of the area occupied by all children of a given node. The returned value depends on the scaling. If Owner.Rotated=True, returns the total height of the children area instead of the width. If the node is collapsed (Expanded=False), then returns 0.

See Also:

TdxOcNode.ChildrenHeight, TdxOcNode.FullWidth

ClientRect method

Returns a node's client area.

```
function ClientRect: TRect;
```

Description

Returns coordinates of the client area of a node, i.e. of the rectangle inside of which text, image and (possible) additional information is displayed. The client area coincides with the contour of the node for rectangle nodes (Shape=shRectangle), and for other node types, it is always less.

Collapse method

Hides all children of a node.

```
procedure Collapse(Recurse: Boolean);
```

Description

Collapses the node, i.e. makes its children invisible on the screen. If Recurse is True, then all the children of a given node are collapsed at all levels.

See Also:

TdxOcNode.Expand, TdxOcNode.Expended

Create method

Creates an instance of TdxOcNode.

```
constructor Create(AOwner: TdxCustomOrgChart);
```

Description

Creates the TdxOcNode object belonging to the AOwner object of TdxCustomOrgChart class. Properties of the object get the following values upon creation:

Owner	AOwner
Width	0 (default)
Height	0 (default)
ChildAlign	caCenter
Color	clNone (default)
Shape	shRectangle
ImageIndex	-1
ImageAlign	AOwner.DefaultImageAlign

Applications usually do not need to call Create directly. For the creation of new nodes, they should use TdxCustomOrgChart methods. Direct Create calls may be required only for an overridden CreateNode method.

See Also:

TdxCustomOrgChart.CreateNode, TdxCustomOrgChart.OnCreateNode

DeleteChildren method

Deletes all children of a node.

procedure DeleteChildren;

Description

Deletes all the children of a given node. The list of children is saved but emptied. To delete children together with the list, use HasChildren = False.

See Also:

TdxCustomOrgChart.Delete, TdxOcNode.HasChildren

Destroy method

Destroys an instance of the TdxOcNode object.

destructor Destroy; override;

Description

Deletes a TdxOcNode object. If the destructor has been called directly (i.e via Free), the object deletes itself from the tree structure calling Owner.Delete.

See Also:

TdxCustomOrgChart.Delete

DisplayRect method

Returns the bounding rectangle for a node.

function DisplayRect: TRect;

Description

Returns coordinates of the area occupied by a given node on the screen. Besides the node image itself, it includes a blank space above (or on the left if Owner.Rotated=True) separating the node from nodes of the previous level.

See Also:

TdxCustomOrgChart.Options, TdxOcNode.ClientRec

Expand method

Shows the children of a node.

procedure Expand(Recurse: Boolean);

Description

Expands the node, i.e. makes its children visible on the screen. If Recurse is True, then all the children of a given node are expanded at all levels.

See Also:

TdxOcNode.Collapse, TdxOcNode.Expanded

Focused method

Indicates whether a node appears to have focus.

function Focused: Boolean;

Description

Returns True, if a node has input focus. If ocFocus mode in Owner.Options is True, then the Focusd node is marked by clHighlight color.

Note: Though tree nodes are not windowed controls and cannot have real input focus, they can be edited by a user when their owner has focus.

See Also:

TdxCustomOrgChart.Options, TdcOcNode.Selected

FullHeight method

Returns the height of a node's "home" area.

function FullHeight: Integer;

Description

Returns the total height of the area occupied by a node with all its children. The returned value depends on scaling. If Owner.Rotated=True, returns the total width of the children area instead of the height. The value of FullHeight changes by expansion and collapse of a node.

See Also:

TdxOcNode.ChildrenHeight, TdxOcNode.FullWidth

FullWidth method

Returns the width of node's "home" area.

function FullWidth: Integer;

Description

Returns the total width of the area occupied by a node with all its children. The returned value depends on scaling. If Owner.Rotated=True, returns the total height of the children area instead of the width. The value of FullWidth changes on expansion and collapse of a node.

See Also:

TdxOcNode.ChildrenWidth, TdxOcNode.FullHeight

GetFirstChild method

Returns the first child node of a tree node.

function GetFirstChild: TdxOcNode;

Description

Call GetFirstChild to refer to the first child of a given node. GetFirstChild returns the same value as Items[0] or nil if the node has no children.

See Also:

TdxOcNode.GetLastChild

GetLastChild method

Returns the last immediate child of a tree node.

function GetLastChild: TdxOcNode;

Description

Call GetLastChild to find the last immediate child of a given node. GetLastChild returns the same value as Items[Count-1] or nil if the node has no children.

See Also:

TdxOcNode.GetFirstChild

GetNext method

Returns the next node after the calling node in a tree.

function GetNext: TdxOcNode;

Description

Returns reference to the next node in a tree data structure, including children and invisible nodes. Use GetNext to iterate all tree nodes, for example:

```
Node := GetFirstNode;
    while Node <> nil do begin
        Node.Color := clGreen;
        Node := Node.GetNext;
    end;
```

See Also:

TdxOcNode.GetNextSibling, TdxOcNode.GetNextViosible, TdcOcNode.GetPrev

GetNextChild method

Returns the next child node after the value.

function GetNextChild(Value: TdxOcNode): TdxOcNode;

Description

Returns reference to a node coming after Value in the children list of a given node. If Value is not an immediate child of a given node, it returns nil.

See Also:

TdxObNode.GetFirstChild, TdxObNode.GetLastChild, TdxObNode.GetPrevChild

GetNextSibling method

Returns the next node in the tree at the same level as the calling node.

function GetNextSibling: TdxOcNode;

Description

GetNextSibling will return the next node, regardless of whether it is visible. If a given node is the last in the children list of its parent, it returns nil.

See Also:

TdxOcNode.GetNext, TdxOcNode.GetNextVisible, TdxOcNode.GetPrevSibling

GetNextVisible method

Returns the next visible node in a tree after the calling node.

function GetNextVisible: TdxOcNode;

Description

Use GetNextVisible to iterate through visible nodes: for example, in selecting nodes by arrow keys. The node is visible on the screen if all its parents are expanded.

See Als:

TdxOcNode.GetNext, TdxOcNode.GetPrevVisible, TdxOcNode.GetNextSibling

GetNodeInfo method

Retrieves the values of a node's properties into the data structure.

```
TdxOcNodeAlign = (caLeft, caCenter, caRight);
TdxOcShape = (shRectangle, shRoundRect, shEllipse, shDiamond);
TdxOcImageAlign = (iaNone, iaLT, iaLC, iaLB, iaRT, iaRC, iaRB,
  iaTL, iaTC, iaTR, iaBL, iaBC, iaBR);
TdxOcNodeInfo = packed record
  Width: Word;
  Height: Word;
  Color: TColor;
  Align: TdxOcNodeAlign;
  Shape: TdxOcShape;
  Index: Smallint;
  IAlign: TdxOcImageAlign;
end;
procedure GetNodeInfo(var AInfo: TdxOcNodeInfo);
```

Description

Fills in the AInfo structure with fields values of the TdxOcNode object. Used by ReadData and WriteData methods. GetNodeInfo is also helpful when it is required to get non-actual values but a user's settings of Width, Height and Color properties.

See Als:

TdxOcNode.ReadData, TdxOcNode.WriteData

GetPrev method

Returns the previous node before the calling node in a tree.

function GetPrev: TdxOcNode;

Description

Returns reference to the previous node in a data structure including children and invisible nodes. If the given node is the first on the 0 level, it returns nil.

See Also:

TdxOcNode.GetNext, TdxOcNode.GetPrevSibling, TdxOcNode.PrevVisible

GetPrevChild method

Returns the previous child node before the value.

function GetPrevChild(Value: TdxOcNode): TdxOcNode;

Description

Returns reference to the node previous to Value in the children list of a given node. If Value is not a direct child of a given node, it returns nil.

See Also:

TdxOcNode.GetFirstChild, TdxOcNode.GetLastChild, TdxOcNode.GetNextChild

GetPrevSibling method

Returns the previous node in a tree at the same level as the calling node.

function GetPrevSibling: TdxOcNode;

Description

Returns the previous sibling node, regardless of whether it is visible. If a given node is the first in the children list of its parent, it returns nil.

See Als:

TdxOcNode.GetNextSibling, TdxOcNode.GetPrev, TdxOcNode. GetPrevVisible

GetPrevVisible method

Returns the previous visible node in the tree after a calling node.

function GetPrevVisible: TdxOcNode;

Description

Returns reference to the foregoing the given node visible node. The node is visible on the screen if all its owners are expanded.

See Also:

TdxOcNode.GetNextVisible, TdxOcNode.GetPrev, TdxOcNode.GetPrevSibling

GetText method

Returns a node's text.

```
function GetText: String; virtual;
```

Description

GetText is a protected implementation of reading the Text property. By default calls the Owner.OnGetText event handler; and if it is not defined, returns the contents of the private field in TdxOcNode object storing text string. Overriding GetText and SetText methods, allows descendants of TdxOcNode to implement another scheme of access to text. Such implementation is also available for applications via the OnGetText and OnSetText events.

See Also:

TdxCustomOrgChart.OnGetText, TdxOcNode.SetText, TdxOcNode.Text

HasAsParent method

HasAsParent returns True if Value is a parent node for a given node.

```
function HasAsParent(Value: TdxOcNode): Boolean;
```

Description

Use HasAsParent to determine whether Value is direct (or indirect) parent of a given node.

See Also:

TdxOcNode.Parent

IndexOf method

IndexOf returns the position of a node's immediate child.

```
function IndexOf(Value: TdxOcNode): Integer;
```

Description

Returns index of Value in the children list of a given node. If Value is not an immediate child of a given node, it returns -1.

See Also:

TdxOcNode, Index, TdxOcNode.Parent

MakeExposed method

```
procedure MakeExposed;
```

Description

The whole node is made visible: calls MakeVisible and in case of necessity runs scrolling of the component in a way to put the node in a visible area of the window.

See Also:

TdxOcNode.MakeVisible

MakeVisible method

Makes a node visible on the screen.

```
procedure MakeVisible;
```

Description

The whole node is made visible: expands all its parents and in case of necessity scrolls the component in a way to place the node in the visible area of a window.

See Also:

TdxOcNode.IsVisible, TdxOcNode.MakeExposed

MoveTo method

Moves a node to another location in the tree.

```
TdxOcNodeAttachMode = (naAdd, naAddFirst, naAddChild, naAddChildFirst,
naInsert);
procedure MoveTo(Dest: TdxOcNode; Mode: TdxOcNodeAttachMode);
```

Description

Moves a node with all its children to a new location in the data structure. Dest sets a node relative to which the location of a given node is determined; when Dest=nil, node moves to the very top level. Mode sets the means by which to move a node:

naAdd - Shifts to the same level as Dest, to the end of the list.

naAddFirst - Shifts to the same level as Dest, to the beginning of the list.

naAddChild - Becomes the last descendant of Dest.

naAddChildFirst - Becomes the first descendant of Dest.

naInsert - Inserted right before Dest, to the same level. Dest cannot be equal to nil.

See Also:

TdxCustomOrgChart.GetNodeAt

ReadData method

Loads a node's properties from the stream.

```
procedure ReadData(Stream: TStream); virtual;
```

Description

Called by reading the TdxCustomOrgChart object from the stream. By default, reads from the stream values of Width, Height, Color, ChildAlign, Shape, ImageAlign properties.

Descendants of TdxOcNode can override ReadData and WriteData methods for reading and storing additional data. The overridden ReadData method must call inherited ReadData first, and only then it can read values of the additional fields from the stream.

```
TMyNode = class(TdxOcNode)
  Prop1: Word;
  Prop2: Integer;
  procedure ReadData(Stream: TStream); override;
  procedure WriteData(Stream: TStream); override;
end;

procedure TMyNode.ReadData(Stream: TStream);
begin
  inherited ReadData(Stream);
  Stream.ReadBuffer(Prop1, SizeOf(Prop1));
```

```
Stream.ReadBuffer(Prop2, SizeOf(Prop2));  
end;
```

See Also:

TdxOcNode.WrietData

SetData method

Sets a value for the Data property.

```
procedure SetData(Value: Pointer); virtual;
```

Description

Set data is a protected implementation of writing the Data property. By default, stores Value to the private field of the TdxocNode object. It may be overridden in order to set reaction to a change of the Data property. The overridden SetData method must always call inherited SetData.

See Also:

TdxOcNode.Data

SetFont method

Sets the font properties for a node.

```
procedure SetFont(Font: TFont); virtual;
```

Description

Called on drawing a node on the screen and also on calling its text editor. By default, calls the Owner.OnSetFont event handler if it is defined. By overriding SetFont, descendants of TdxOcNode can set individual fonts for any node.

See Also:

TdxCustomOrgChart.OnSetFont

SetText method

Stores the new value of a node's text.

```
procedure SetText(const Value: String); virtual;
```

Description

SetText is a protected implementation of writing the Text property. By default, calls the Owner.OnSetText event handler; and if it is not defined, stores the Value argument to the private field of the TdxOcNode object keeping text string. By overriding GetText and SetText methods, descendants of the TdxOcNode can implement another scheme of access to text. Such implementation is also available for applications via the OnGetText and OnSetText events.

See Also:

TdxCustomOrgChart.OnSetFont, TdxCustomOrgChart.OnSetText, TdxOcNode.GetText

WriteData method

Stores a node's properties in the stream.

```
procedure WriteData(Stream: TStream); virtual;
```

Description

Called while storing the TdxCustomOrgChart object to the stream. By default, stores values of Width, Height, Color, ChildAlign, Shape, ImageIndex and ImageAlign properties to the stream. Descendants of TdxOcNode can override ReadData and WriteData methods for reading and storing additional data. The overridden WriteData method must call inherited WriteData first, and only then can it record values of the additional fields to the stream.

```
TMyNode = class(TdxOcNode)
    Prop1: Word;
    Prop2: Integer;
    procedure ReadData(Stream: TStream); override;
    procedure WriteData(Stream: TStream); override;
end;

procedure TMyNode.WriteData(Stream: TStream);
begin
    inherited WriteData(Stream);
    Stream.WriteBuffer(Prop1, SizeOf(Prop1));
    Stream.WriteBuffer(Prop2, SizeOf(Prop2));
end;
```

TdxDbOrgChart

Unit

DxDbOrgC

Description

TdxDbOrgChart is fully functional windowed data-aware component, intended for the display and editing of hierarchical tree data structures. The component connects to a database via a TDataSet class component, from which it obtains all necessary data for its tree and individual node formation. The hierarchy of nodes is defined via the database fields KeyField and ParentField while other (optional) fields may contain properties of separate nodes. Any changes made by a user are immediately displayed in the database and any changes to the database implemented by your program are immediately displayed in TdxDbOrgChart component.

TdxDbOrgChart Properties

In TdxDbOrgChart

ChAlignField	public/read/write
ChAlignFieldName	public/read/write
ColorField	public/read/write
ColorFieldName	public/read/write
DataSet	public/read/write
DataSource	published/read/write
HeightField	public/read/write
HeightFieldName	public/read/write
ImageField	public/read/write
ImageFieldName	published/read/write
ImAlignField	public/read/write
ImAlignFieldName	public/read/write
KeyField	public/read/write
KeyFieldName	published/read/write
ParentField	public/read/write
ParentFieldName	published/read/write
ShapeField	public/read/write
ShapeFieldName	public/read/write
TextField	public/read/write
TextFieldName	published/read/write
WidthField	public/read/write
WidthFieldName	public/read/write

Derived from TdxCustomOrgChart

BorderStyle	published/read/write
Count	public/read/write
DefaultImageAlign	published/read/write
DefaultNodeHeight	published/read/write
DefaultNodeWidth	published/read/write
Editing	public/read/write
EditMode	published/read/write
Images	published/read/write
IndentX	published/read/write
IndentY	published/read/write

Items	public/read/write
LeftEdge	public/read/write
LineWidth	published/read/write
Options	published/read/write
Rotated	published/read/write
Selected	public/read/write
TopEdge	public/read/write
Zoom	published/read/write

ChAlignField property

[read/write](#)

ChAlignField is a data field for the TdxDbOcNode.ChildAlign property.

property ChAlignField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the ChildAlign property of tree nodes. Can be changed via a setting of ChAlignFieldName.

See Also:

TdxDbOrgChart.ChAlignFieldName, TdxOcNode.ChAlign

ChAlignFieldName property

[read/write](#)

ChAlignFieldName is the name of field ChAlignField.

property ChAlignFieldName: String;

Description

Sets the name of ChAlignField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the ChAlignField.

See Also:

TdxDbOrgChart.ChAlignField

ColorField property

[read/write](#)

ColorField is a data field for TdxDbOcNode.Color property.

property ColorField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the Color property of tree nodes. Can be changed via a setting of ColorFieldName.

See Also:

TdxDbOrgChart.ColorFieldName, TdxOcNode.Color

ColorFieldName property

[read/write](#)

ColorFieldName is the name of field ColorField.

property ColorFieldName: String;

Description

Sets the name of ColorField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the ColorField.

See Also:

TdxDbOrgChart.ColorField

DataSet property

read/write

Specifies the dataset which contains the data for a tree structure.

property DataSet: TDataSet;

Description

The DataSet property contains a reference to the TDataSet component, determining (via DataSource) a database connected with an OrgChart component.

See Also:

TdxDbOrgChart.DataSource

DataSource property

read/write

DataSource identifies the link to the dataset from which a component loads its data.

property DataSource: TDataSource;

Description

Set DataSource to the TDataSource object that links to the dataset from which the TdxDbOrgChart component should fetch its data. DataSource allows the component to read from, write to, and navigate around a dataset, while maintaining record concurrency with other data-aware objects that use the same data source.

HeightField property

read/write

HeightField is a data field for TdxDbOcNode.Height property.

property HeightField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the Height property of tree nodes. Can be changed via setting of HeightFieldName.

See Also:

TdxDbOrgChart.HeightFieldName, TdxOcNode.Height

HeightFieldName property

read/write

HeightFieldName is the name of field HeightField.

property HeightFieldName: String;

Description

Sets the name of HeightField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the HeightField.

See Also:

TdxDbOrgChartHeightField

ImageField property**read/write**

ImageField is a data field for TdxDbOcNode.ImageIndex property.

property ImageField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the ImageIndex property of tree nodes. Can be changed via a setting of ImageFieldName.

See Also:

TdxDbOrgChart.ImageFieldName, TdxOcNode.ImageIndex

ImageFieldName property**read/write**

ImageFieldName is the name of field ImageField.

property ImageFieldName: String;

Description

Sets the name of ImageField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the ImageField.

See Also:

TdxDbOrgChart.ImageField

ImAlignField property**read/write**

ImAlignField is a data field for TdxDbOcNode.ImAlign property.

property ImAlignField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the ImageAlign property of tree nodes. Can be changed via a setting of ImAlignFieldName.

See Also:

TdxDbOrgChart.ImAlignFieldName, TdxOcNode.ImAlign

ImAlignFieldName property**read/write**

ImAlignFieldName is the name of field ImAlignField.

property ImAlignFieldName: String;

Description

Sets the name of ImAlignField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the ImAlignField.

See Also:

TdxDbOrgChart.ImAlignField

KeyField property**read/write**

KeyField contains a key value for tree nodes corresponding to a given database record.

property KeyField: TField;

Description

Contains a reference to the required field of a connected database setting the Key property of tree nodes. Can be changed via setting of KeyFieldName. This field can be of any type, though it is usually defined as an integer. Values in KeyField field must be unique for each record and must not be empty. Key value duplication for several records will force only a single node to be built. Records with blank key value are ignored.

See Also:

TdxDbOcNode.Key, TdxDbOrgChart.KeyFieldName

KeyFieldName property

read/write

KeyFieldName is the name of field KeyField.

property KeyFieldName: String;

Description

Sets the name of KeyField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the KeyField nor will it have a database connection.

See Also:

TdxDbOrgChart.KeyField

ParentField property

read/write

ParentField determines the parent of a tree node corresponding to a given database record.

property ParentField: TField;

Description

Contains a reference to the required field of a connected database setting the ParentKey property of tree nodes. Can be changed via setting of ParentFieldName. Parent Field type must match with KeyField type. This field determines the parent of a node connected with a given record. Blank ParentField field means, that a given node is at the top level of the hierarchy without a parent. The same result can be achieved by setting the ParentField value equal to KeyField (the node refers to itself). Records where the node refers to its child are by definition invalid and as a result are ignored.

See Also:

TdxDbOcNode.ParetnKey, TdxDbOrgChart.ParentFieldName, TdxOcNode.Parent

ParentFieldName property

read/write

ParentFieldName is the name of field ParentField.

property ParentFieldName: String;

Description

Sets the name of ParentField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the ParentField nor will it have a database connection.

See Also:

TdxDbOrgChart.ParentField

ShapeField property**read/write**

ShapeField is a data field for TdxDbOcNode.Shape property.

property ShapeField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the Shape property of tree nodes. Can be changed via setting of ShapeFieldName.

See Also:

TdxDbOrgChart.ShapeFieldName, TdxOcNode.Shape

ShapeFieldName property**read/write**

ShapeFieldName is the name of field ShapeField.

property ShapeFieldName: String;

Description

Sets the name of ShapeField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the ShapeField.

See Also:

TdxDbOrgChart.ShapeField

TextField property**read/write**

TextField is a data field for TdxDbOcNode.Text property.

property TextField: TField;

Description

Contains a reference to the optional field of a connected database setting the Text property of tree nodes. Can be changed via setting of TextFieldName. This field may be of any type, though the TStringField type is more frequently used.

See Also:

TdxDbOrgChart.TextFieldName, TdxOcNode.Text

TextFieldName property**read/write**

TextFieldName is the name of field TextField.

property TextFieldName: String;

Description

Sets the name of TextField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the TextField.

See Also:

TdxDbOrgChart.TextField

WidthField property**read/write**

WidthField is a data field for TdxDbOcNode.Width property.

property WidthField: TField;

Description

Contains a reference to the optional integer field of a connected database setting the Width property of tree nodes. Can be changed via setting of WidthFieldName.

See Also:

TdxDbOrgChart.WidthFieldName, TdxOcNode.Width

WidthFieldName property

read/write

WidthFieldName is the name of field WidthField.

property WidthFieldName: String;

Description

Sets the name of WidthField, defined in a connected database. If you set a name which does not exist in a database, the component will not have the WidthField.

See Also:

TdxDbOrgChart.WidthField

TdxDbOrgChart Methods

In TdxDbOrgChart

Active	public
Create	public
CreateEditor	protected
CreateNode	protected
Delete	public
Destroy	public
DoChange	protected
DoChanging	protected
InternalAdd	protected
InternalMoveTo	protected
NodeChanged	protected
ShowEditor	public

Derived from TdxCustomOrgChart

CreateParams	DbClick	DragOver	DrawNode
KeyDown	KeyPress	MouseDown	MouseMove
Paint	Loaded	Add	AddChild
AddChildFirst	AddFirst	AssignData	BeginUpdate
Clear	DoZoom	DragDrop	EndUpdate
FullCollapse	FullExpand	FullHeight	FullWidth
GetFirstNode	GetHitTestsAt	GetNodeAt	HideEditor
Insert	Invalidate	IsZoomed	SetLeftTop
ShowEditor			

Active method

Active reports whether a component is active.

function Active: Boolean;

Description

Returns True, if the component is active, i.e. the connected database is opened and the fields KeyField and ParentField are assigned.

See Also:

TdxDbOrgChart.KeyField, TdxDbOrgChart.ParentField

Create method

constructor Create(AOwner: TComponent); override;

Description

Overrides TdxCustomOrgChart.Create. Having called inherited Create creates a TdxOcDataLink object and internal list ordered according to the key values.

See Also:

TdxCustomOrgChart.Create

CreateEditor method

Creates the in-place text editor for tree nodes.

function CreateEditor: TdxOcInplaceEdit; override;

Description

Overrides TdxCustomOrgChart.CreateEditor. If the TextField is active, restricts the length of the edited text to no more than the length of the TextField.

See Also:

TdxCustomOrgChart.CreateEditor

CreateNode method

Creates a new tree node.

function CreateNode: TdxOcNode; override;

Description

Overrides TdxCustomOrgChart.CreateNode. Creates an object of TdxDbOcNode class and sets its Key and ParentKey properties to null.

See Also:

TdxCustomOrgChart.CreateNode, TdxOcNode.Key, TdxOcNode.ParentKey

Delete method

Removes a node from the tree structure.

procedure Delete(ANode: TdxOcNode); override;

Description

Overrides TdxCustomOrgChart.Delete. Before calling inherited Delete, removes the connected record with ANode from the database.

See Also:

TdxCustomOrgChart.Delete

Destroy method

Destroys an instance of TdxDbOrgChart.

destructor Destroy; override;

Description

Do not call `Destroy` directly in an application. Instead, call `Free`. `Free` verifies that the `TdxDBOrgChart` object is not already freed and only then calls `Destroy`. `Destroy` frees up memory associated with data handling that was allocated in the constructor, before calling the inherited destructor.

See Also:

TdxCustomOrgChart.Destroy

DoChange method

`DoChange` notifies that a selection has changed from one tree node to another.

procedure `DoChange(Node: TdxOcNode); override;`

Description

Overrides `TdxCustomOrgChart.DoChange`. Sets the current record of the database by `Key` property of the `Node`, then calls inherited `DoChange` in a manner as to leave the `OnChange` event available for applications.

See Also:

TdxCustomOrgChart.OnChange, TdxCustomOrgChart.DoChange

DoChanging method

`DoChanging` notifies that a selection is about to be changed from one tree node to another.

procedure `DoChanging(Node: TdxOcNode; var Allow: Boolean); override;`

Description

Overrides `TdxCustomOrgChart.DoChanging`. First calls inherited `DoChanging` for `OnChanging` event processing. If after that `Allow = True`, sets the current record of the database by the `Key` property of a `Node`. If it fails, disables selection of the `Node` and returns the database cursor to the previous location.

See Also:

TdxCustomOrgChart.OnChanging, TdxCustomOrgChart.DoChanging

InternalAdd method

Adds a new node to the tree structure.

function `InternalAdd(AParent: TdxOcNode; Ptr: Pointer; AIndex: Integer): TdxOcNode; override;`

Description

Overrides `TdxCustomOrgChart.InternalAdd`. With inactive databases, it immediately returns `nil`. Otherwise calls inherited `InternalAdd` and sets `Key` (creating a new key) and `ParentKey` (equal `AParentKey`) properties for the created node. After this, it adds a record to the database and fills in the fields `KeyField`, `ParentField` and `OrderField` (if defined). Finally, where necessary, alters the `OrderField` (or `KeyField`) values for all nodes, following the newly created one in the `AParent` list of children.

See Also:

TdxCustomOrgChart.InternalAdd

InternalMoveTo method

Moves a tree node to new position in the tree structure.

```
procedure InternalMoveTo(AParent, ANode: TdxOcNode;  
AIndex: Integer); override;
```

Description

Overrides TdxCustomOrgChart.InternalMoveTo. Calls inherited InternalMoveTo and alters the ANode.ParentKey property indicating changes in the database. Where necessary, alters the OrderField (or KeyField) value of the ANode and of all following it in the AParent children list.

See Also:

TdxCustomOrgChart.InternalMoveTo

NodeChanged method

Notifies that one or more properties of a node have changed.

```
procedure NodeChanged(ANode: TdxOcNode); override;
```

Description

Overrides TdxCustomOrgChart.NodeChanged. On changing the properties of ANode, updates the connected record of the database.

See Also:

TdxCustomOrgChart.NodeChanged

ShowEditor method

Starts the text editing for a selected node.

```
procedure ShowEditor; override;
```

Description

Call ShowEditor to start editing the text of a selected node. Do not call ShowEditor for read only datasets.

See Also:

TdxCustomOrgChart.ShowEditor

TdxDbOrgChart Events

In TdxDbOrgChart

OnLoadNode	published
OnNewKey	published

Derived from TdxCustomOrgChart

OnCreateNode	OnChange	OnChanging	OnCollapsed
OnCollapsing	OnDeletion	OnExpanded	OnExpansion
OnEditing	OnEdited	OnSetFont	OnDrawNode

OnGetText OnSetText

OnLoadNode event

OnLoadNode occurs when a node has been created and its properties have been read from the dataset.

TdxOcEvent = **procedure**(Sender: TObject; Node: TdxOcNode) of object
property OnLoadNode: TdxOcEvent;

Description

Use the OnLoadNode event to perform specific operations after the node has been created based on the data in a linked dataset.

```
procedure TForm1.ChartLoadNode(Sender: TObject; Node: TdxOcNode);
begin
    if Node.Parent=nil then Node.Shape := shRoundRect;    // by Node's
properties
    if Chart.DataSet.FieldByName('Flag').AsBoolean then
        Node.Color := clRed;        // by contents of database
end;
```

On calling OnLoadNode, the current record in the database corresponds to the Node and the event handler must not change its location (Do not use Locate, First, Last and MoveBy methods of the dataset or use bookmark to return to the node that corresponds with the Node).

OnNewKey event

OnNewKey occurs when a new key value is required.

TdxOcNewKeyEvent = **procedure**(Sender: TObject; MaxValue: Variant;
 var KeyValue: Variant) of object;
property OnNewKey: TdxOcNewKeyEvent;

Description

Called in all cases when it is required to create a new key: on adding new nodes, and also on changing the tree structure by Add... Insert and MoveTo methods. The event handler must assign to a KeyValue a variable value bigger than of MaxValue. New keys are created automatically if the KeyField is an integer type. In this case to define the OnNewKey event handler is not required. If the KeyField type differs from an integer (for example string), the OnNewKey event handler must be defined.

```
procedure TForm1.ChartNewKey(Sender: TObject; MaxValue: Variant;
    var KeyValue: Variant);
begin
    if VarIsNull(MaxValue) then KeyValue := '00000000'
    else begin
        KeyValue := IntToStr(StrToInt(MaxValue)+1);
        while Length(KeyValue) < 8 do KeyValue := '0' + KeyValue;
    end;
end;
```

TdxDbOcNode

Unit

DxDbOrgC

Description

TdxDbOcNode is a class, describing a separate node in the TdxDbOrgChart data structure, to which corresponds a separate record in connection with a TdxDbOrgChart database. TdxDbOcNode is a direct descendant of TdxOcNode and inherits all its properties and methods.

TdxDbOcNode Properties

In TdxDbOcNode

Key	public/read/write
ParentKey	public/read/write

Derived from TdxOcNode

ChildAlign	public/read/write
Color	public/read/write
Count	public/read/write
Data	public/read/write
Deleting	public/read/write
Expanded	public/read/write
HasChildren	public/read/write
Height	public/read/write
ImageAlign	public/read/write
ImageIndex	public/read/write
Index	public/read/write
IsVisible	public/read/write
Items	public/read/write
Level	public/read/write
Owner	public/read/write
Parent	public/read/write
Selected	public/read/write
Shape	public/read/write
Text	public/read/write
Width	public/read/write

Key property

read/write

Represents a KeyField value for a tree node.

property Key: Variant;

Description

Contains the key value of a node equal to the value of KeyField. Automatically created on creation of a node and (if KeyOrder is True) automatically altered upon changes to its location. For nodes with common parent the Key values, determines the ordering in the structure, if the OrderField is not defined.

See Also:

TdxDbOcNode.Parent Key, TdxDbOcNode.KeyField

ParentKey property**read/write**

Represents the ParentField value for a tree node.

property ParentKey: Variant;

Description

Run-time and read only. Use ParentKey to determinate the Value of the Parent field, which is associated with a TdxDBOcNode.

See Also:

TdxDbOcNode.Key, TdxDbOcNode.ParentField

TdxDbOcNode Methods

In TdxDbOcNode

SetText	protected
DbOwner	public
ReadData	protected

Derived from TdxOcNode

GetText	SetData	SetFont	WriteData
Create	Destroy	ChildrenHeight	ChildrenWidth
ClientRect	Collapse	DeleteChildren	DisplayRect
Expand	Focused	FullHeight	FullWidth
GetFirstChild	GetLastChild	GetNext	GetNextChild
GetNextSibling	GetNextVisible	GetNodeInfo	GetPrev
GetPrevChild	GetPrevSibling	GetPrevVisible	HasAsParent
IndexOf	MakeExposed	MakeVisible	MoveTo

SetText method

SetText implements writing the node's Text property.

```
procedure SetText(const Value: String); override;
```

Description

Overrides TdxOcNode.SetText. If the Owner.TextField exists and the length of the string Value exceeds the length of this field, then the string is shortened to an acceptable length and then inherited SetText is called. OnGetText and OnSetText events are still available in TdxDbOrgChart, but it is not recommended to use them when TextField is active.

See Also:

TdxCustomOrgChart.OnGetText, TdxCustomOrgChart.OnSetText, TdxOcNode.SetText

DbOwner method

```
function DbOwner: TdxDbOrgChart;
```

Description

Returns reference to the TdxDbOrgChart component, possessing the given node. The returned value coincides with the value of the Owner property but has the type of TdxDbOrgChart.

ReadData method

```
procedure ReadData(Stream: TStream); override;
```

Description

Overrides TdxOcNode.ReadData. During the call of inherited ReadData bans access to database, later it allows it again and calls Owner.NodeChanged to indicate changes into database.

See Also:

TdxDbOrgChart.NodeChanged, TdxOcNode.ReadData

TdxOcDataLink

Unit

DxDbOrgC

Description

Objects of the TdxOcDataLink class are generated on creation of TdxDbOrgChart to enable reaction of the TdxDbOrgChart component to changes in connection with the component database.

TdxOcDataLink Methods

ActiveChanged	protected
DataSetChanged	protected
DataSetScrolled	protected

ActiveChanged method

ActiveChanged responds to changes in the DataSet.Active property.

procedure ActiveChanged; override;

Description

Called on changing the DataSet.Active property, on opening and closing of a database in particular. Additionally, TdxDbOrgChart calls ActiveChanged on any changes to field names: KeyFieldName, TextFieldName and other properties.

ActiveChanged clears data structures by calling Clear and then performs searching of fields in a database by their names. Further, if a component is active, then DataSetChanged is called to update the data.

Changes to names of any field with an open database leads to complete updating of the TdxDbOrgChart component with scanning of the table. For this reason, it is best to perform this operation with a closed database if the setting of several fields is required at once.

```
procedure TForm1.FormCreate(Sender: TObject);
begin
  with Chart do begin
    DataSet.Close;
    WidthFieldName := 'Width';
    HeightFieldName := 'Height';
    ColorFieldName := 'Color';
    ShapeFieldName := 'Shape';
    DataSet.Open;
  end;
end;
```

See Also:

TdxOcDataLink.DataSetChanged

DataSetChanged method

DataSetChanged responds to changes in a dataset.

procedure DataSetChanged; override;

Description

Called on any changes of data in the DataSet component. The TdxOcDataLink.DataSetChanged ignores the message if the database is in an edit state (dsEdit or dsInsert). In all other cases, the TdxDbOrgChart component performs updating of its data by reading the connected DataSet component.

DataSetScrolled method

DataSetScrolled allows a response to scrolling the representation of the dataset within the data-aware object.

procedure DataSetScrolled(Distance: Integer); override;

Description

Called on changing the location of the current record in a database. By this the TdxDbOrgChart component selects (Selected) a node connected with the current record. If the current record is incorrect, the component will not have a selected node.

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